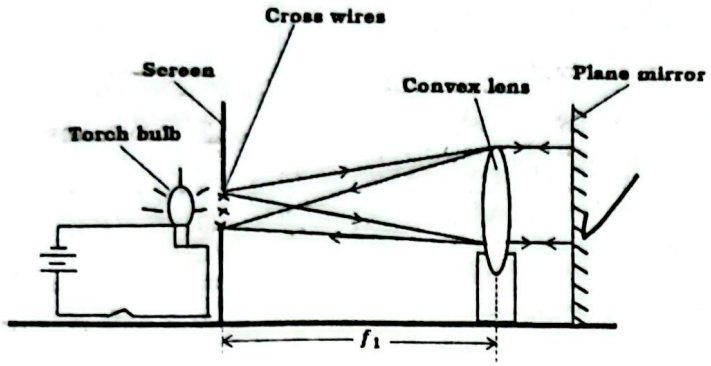
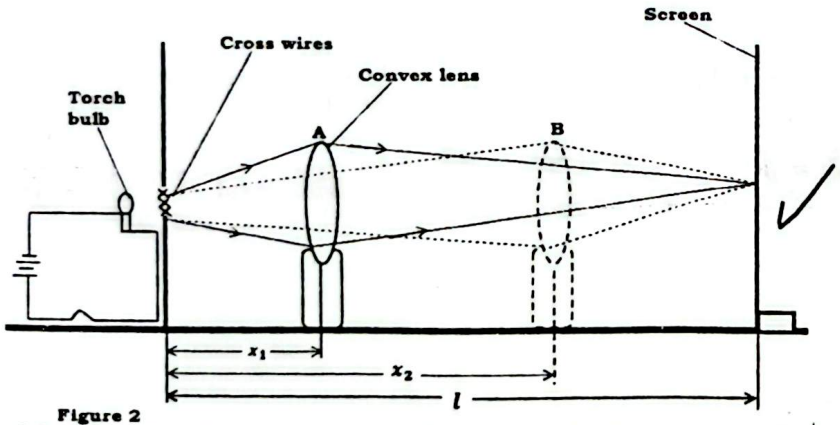
tem 1

BASIS	Remarks	Score
A. PLANNING FOR THE EXPERIMENT (PE) • **Aim** An investigation to determine the power of a convex lens by varying the distance between the illuminated object (cross wires) and the screen and measuring the corresponding object distances when the image is magnified and when diminished respectively so as to assess the suitability of the lens for use in the projection system.	- **Correct statement** - **Relationship between variables** - **Purpose of the investigation**	**01** **01** **01**
• **Variables** Independent variables. Distance between the screen with illuminated cross wires (object) and the screen (image). - Dependent variables: , object distances (when the image is magnified and when diminished), displacement of the convex lens - Controlled variables: Light intensity in the room, nature of the convex lens	- **At least one correct example of each type of variable stated** - **@ 1 score**	**03**
• **Hypothesis** The power of the sourced convex lens is in the range 9.0 diopters and 12.0	- **Correct hypothesis/prediction stated**	**01**
Key points = 07	**Sub total**	**= 07**

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BASIS	Remarks	scores
B. METHODS OF THE EXPERIMENT (ME)		
Method 1		
	- Workable diagram drawn that can be used to obtain results/values.	01
(a) The apparatus were arranged as shown in the set up in **figure 1** above.		
(b) The position of the convex lens was adjusted until a clear image of the cross wires was focused on the screen. ✓		
(c) The distance, f_1 between the lens and the screen was measured using a metre rule and recorded in metres. ✓ ✓ ✓	- Steps followed stated with coherence	
(d) The value of P_1 was calculated from the expression: $P_1 = \frac{1}{f_1}$ ✓ coherence ✓		06
Method 2		
 Figure 2		
(a) The apparatus were arranged as shown in the set up in **figure 2** above.		
(b) Distance, l, between the illuminated cross wires and the screen was adjusted to l = 45.0cm ✓		

(c) The convex lens was placed close to the illuminated cross wires and its position was adjusted until a clear magnified image of the cross wires was formed on the screen when the lens was at position A. (d) The distance x_1 was measured using a metre rule and recorded. (e) The convex lens was moved further towards the screen until a clear diminished image of the cross wires was formed on the screen when the lens was at position B. (f) The distance, x_2 was measured and recorded. (g) The steps (b) to (e) were repeated for other values of l = 55.0, 65.0, 75.0, 85.0 and 95.0cm. (h) The results were tabulated including values of l^2, $d = (x_2 - x_1)$, d^2 and $(l^2 - d^2)$ (i) A graph of $(l^2 - d^2)$ against l was plotted. (j) The slope, S of the graph was determined. (k) The value of f_2 was calculated from the expression: $f_2 = \frac{S}{4}$ and expressed in metres. (l) The value of P_2 was calculated from the expression: $P_2 = \frac{1}{f_2}$ (m) The power, P of the convex lens was determined from the expression $P = \frac{P_1+P_2}{2}$ **Coherence**	- **Workable diagram drawn that can be used to obtain results/values.** - **Steps followed stated with coherence**	**01** **14**
Errors and precautions • Parallax errors when measuring object distances using a metre rule. This can be mitigated by positioning the eye perpendicularly to the scale of the metre rule. • Uncertainty in obtaining clear/sharp images on the screen. This can be mitigated by making several observations of the images on the screen for every value of the distance l between the screen and the illuminated object	- **At least one possible source of errors stated and its corresponding mitigation/ precaution**	**01**
Key points = 23	**Sub total**	**= 23**

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$f_1 = 9.8cm = 0.098m$

$l(cm)$	$x_1(cm)$	$x_2(cm)$	$l^2(cm^2)$	$d(cm)$	$d^2(cm^2)$	$(l^2 - d^2)(cm^2)$
45.0 55.0 65.0 75.0 85.0 95.0	• 1 d.p • Increasing trend	• 1d.p • Increasing trend	• 0 dp • Increasing trend			
Key points = 18						

C. DATA ANALYSIS AND INTERPRETATION (AI)	Remarks	Scores
• Title of the graph: A graph of $(l^2 - l^2)$ against l	- Written correctly. - No units	01
• perpendicular axes drawn and labelled: VA: $(l^2 - d^2)(cm)$ HA: l (cm)	- Perpendicular axes drawn with arrows - @ 1 score	02
• Uniform scales	@ 1 score	02
• - Starting value for the vertical axis sated/indicated -Starting value for the horizontal axis stated/indicated		02
• At least Five correctly plotted points. -	@ 1 score	05
• Line of best fit drawn.	- provided at least 4 points are correctly plotted)	01
• Coordinates correctly and accurately read to the number of d.ps of one small box on the respective axis.	@ 1 score	04

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• Slope, S value correctly calculated with unit ✓	- Provided all the values in the coordinates are correct - S = (36 – 48)cm Or S = (0.36 – 0.48)m	01
• Calculation of f_2: Correct substitution in $f = \frac{S}{4}$ ✓ Value of f_2 calculated with unit ✓	f_2 = (9.0 – 12.0)cm or f_2 = (0.090 – 0.12)m	02
• Calculation of P_1: Correct substitution in $P = \frac{1}{f_1}$ ✓ Value of P_1 calculated with unit ✓	P_1 = (9.0 – 12.0)D	02
• Calculation of P_2: Correct substitution in $P = \frac{1}{f_2}$ ✓ Value of P_2 calculated with unit ✓	P_2 = (9.0 – 12.0)D	02
• Calculation of P Correct substitution in $P = \frac{P_1+P_2}{2}$ ✓ Value of P calculated with unit. ✓	P = (9.0 – 12.0)D	02
Key points = 26	**Sub total**	**= 26**
D. CONCLUSION AND RECOMMENDATION		01
• The power of the sourced convex lens is ---(value stated) which is in the range of 9.0D to12.0D. ✓	Accept a negative statement of the conclusion, provided there is correspondence.	01
• Therefore the sourced convex lens is suitable for use in the projection system. ✓	Accept a negative statement of the recommendation, provided there is correspondence	
Key points = 02	**Sub total**	**= 02**
Total number of key scores = 76	**Grand total**	**76**

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Item 2

BASIS	Description	Scores
A. DATA PRESENTATION (DP)		
Method 1		
• Value of the breadth, b measured and recorded with units to 1dp/2dp in cm or 3dp/4dp in m ✓	b = (2.00 – 2.80)cm OR b = (0.0200 – 0.0280)m	01
• Value of the thickness, t measured and recorded with units to 2dp in mm or 1dp/2dp in cm or 3dp/4dp in m ✓	t = (5.00 – 8.00)mm OR t = (0.50 – 0.80)cm OR t = (0.005 – 0.008)m	01
• Value of the initial position, P_0 of the pointer read and recorded with unit to 1dp in cm or 3dp in m ✓		01
• Value of the new position, P_1 of the pointer read and recorded with unit to 1dp in cm or 3dp in m ✓		01
• Value of the depression, d calculated and recorded with units to 3dp. ✓		01
• Calculation of Y_1. - Correct substitution in $Y = \frac{4gl}{1} \frac{m^2}{bt^3d}$ ✓ - Correctly calculated value of Y_1 with units. ✓		02
Method 2 M(kg) ✓ | t(s) ✓ | T(s) ✓ | $T^2(s^2)$ ✓ 0.050 0.100 0.150 0.200 0.250 ✓ 0.300 - SC: 1dp - Increasing trend - SW: 2dp ✓ - Increasing trend ✓ - SC: 3dp if largest $t < 20.0$ - 2dp if largest $t \geq 20.0$ - SW: 4dp if largest $t < 20.00$ - 3dp if largest $t \geq 20.00$ ✓ - SC: 3dp if largest $t < 20.0$ - 2dp if largest $t \geq 20.0$ - SW: 4dp if largest $t < 20.00$ 3dp if largest $t \geq 20.00$ ✓	- Columnar table of results with each column labelled correctly and corresponding unit symbols in brackets @ 1 score - At least 5 values of M and corresponding realistic values of t - At least 4 realistic values of t recorded Trend = 1 score Accuracy = 1 score - At least 4 values of T calculated accurately. - At least 4 values of T^2 calculated accurately	04 01 02 01 01
• **Possible sources of errors**	- One possible source of errors and the	01

. Parallax errors when reading the positions of the pointer. This is mitigated by positioning the eye perpendicularly to the scale of the vertical half metre rule/metre rule ✓	corresponding mitigation.	
Key points = 17	**Sub total**	**=17**

A. DATA ANALYSIS AND INTERPRETATION (AI)	**Remarks**	**scores**
• Value of Y_1 - Correct Substitution in $Y = \frac{4gl\ m^2}{1\ bt^3d}$ ✓ - Value of Y_1 correctly calculated ✓		**02**
• Title of the graph: A graph of T^2 against M ✓	- **Written correctly, no units**	**O1**
• Perpendicular axes drawn and labelled; VA: $T^2(s^2)$ ✓ HA: M(kg) ✓	**Perpendicular axes drawn with arrows**	**02**
• - Starting value for the vertical axis. ✓ - Starting value for the horizontal axis. ✓		**02**
• Uniform scales ✓✓		**02**
• At least 5 correctly plotted points ✓✓✓✓✓	**@ POINT = 1 score**	**05**
• Line of best ✓		**01**
• Correctly and accurately read coordinates for the slope calculation. (read to the accuracy of the representation of 1 small box) (@ 1 score) ✓✓✓✓	**@ value of a coordinate = 1 score**	**04**
• Correctly calculated value of the slope, S <u>with</u> unit. ✓	- **Provided all the coordinates read from the graph are correct** - **S = (0.0070 – 0.150)mkg^{-1}.** - **3dp or 4dp**	**01**

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• Value of, Y_2. - Correct substitution in $Y = \frac{16\pi l^2 s^3}{5bt^3}$ ✓ - Value of Y_2 with unit. ✓	$Y_2 = 9.0 \times 10^9 Pa - 15 \times 10^{10} Pa$	02
• Value of Y (Average value of Y_1 and Y_2) - Correct substitution in $Y = \frac{Y_1+Y_2}{2}$ ✓ - Value of Y with unit. ✓	$Y = 9.0 \times 10^9 Pa - 15 \times 10^{10} Pa$	02
Key points = 24	**Sub total**	**= 24**

B. CONCLUSION AND RECOMMENDATION (CR)	**remarks**	**scores**
• The Young's modulus of the wood is ...(any correctly obtained value) ✓ • Since the Young's modulus of the wood is in the range of $9.0X10^9Pa$ to $15X10^9Pa$, then the shelves made out of it will not bend excessively ✓	Accept a negative statement of the conclusion, provided there is correspondence. Accept a negative statement of the recommendation, provided there is correspondence	01 01
Key points = 04	**Sub total**	**= 02**
Total number of scores	**Grand total**	**= 43**

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